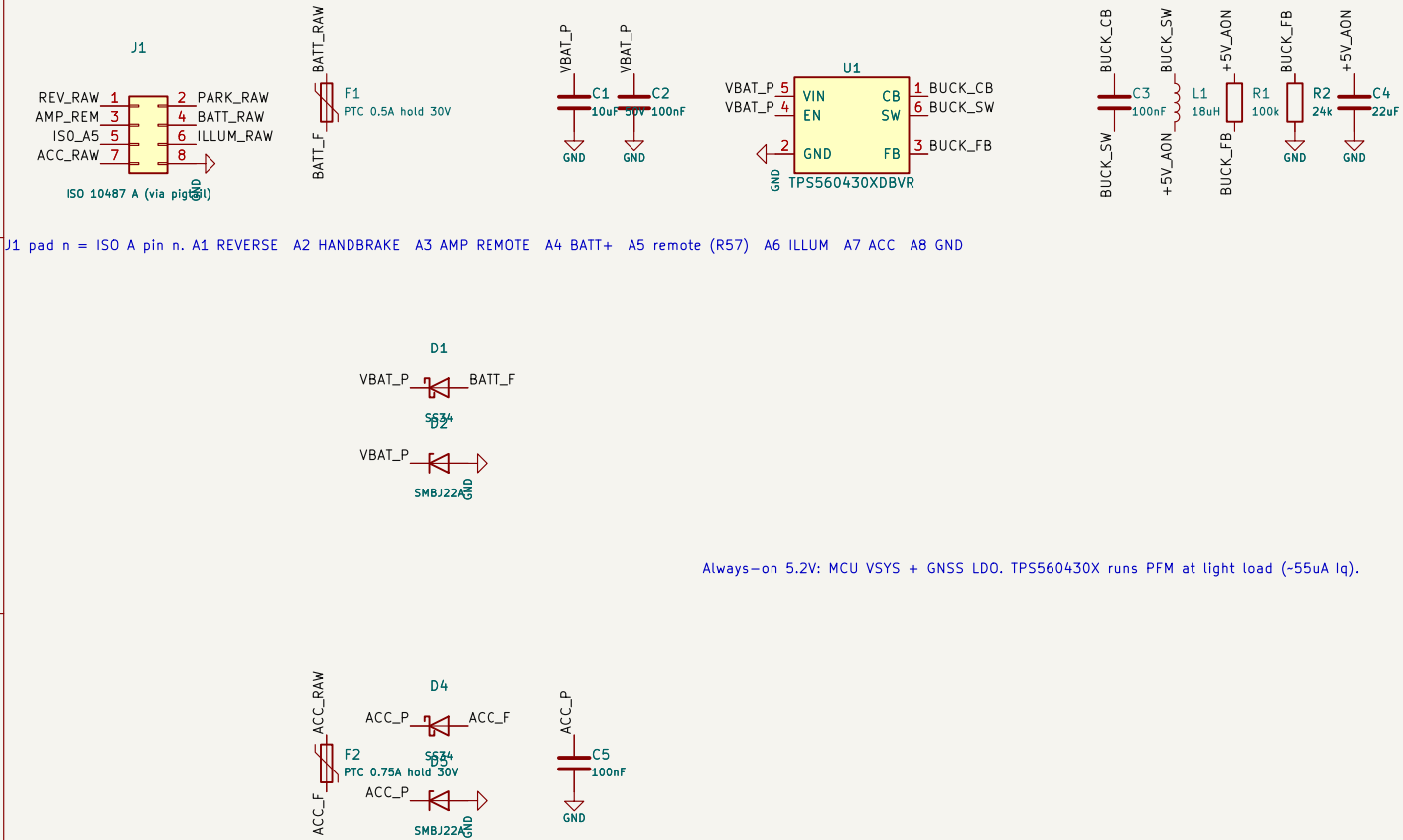


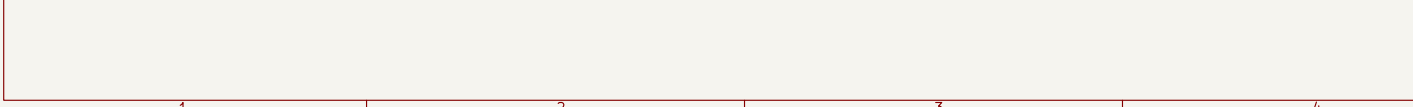
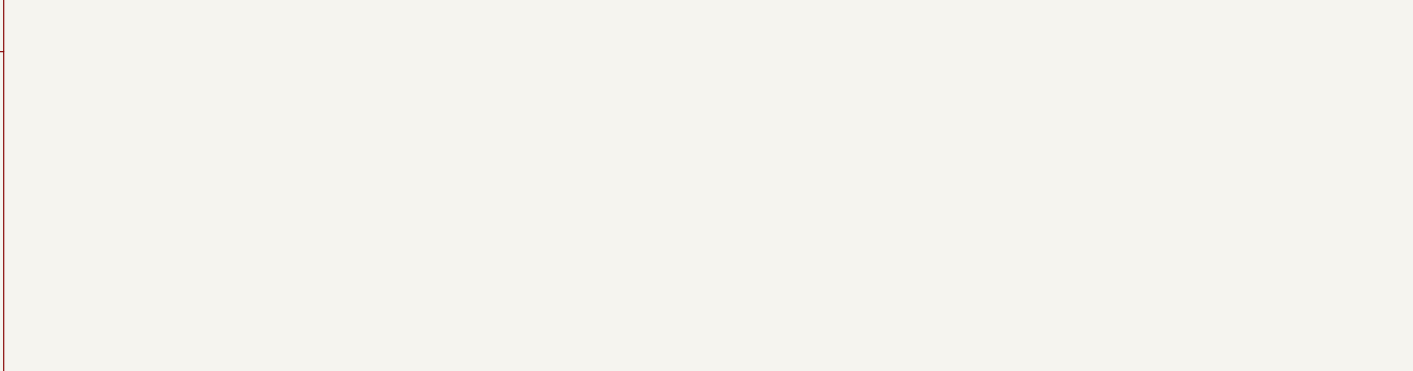
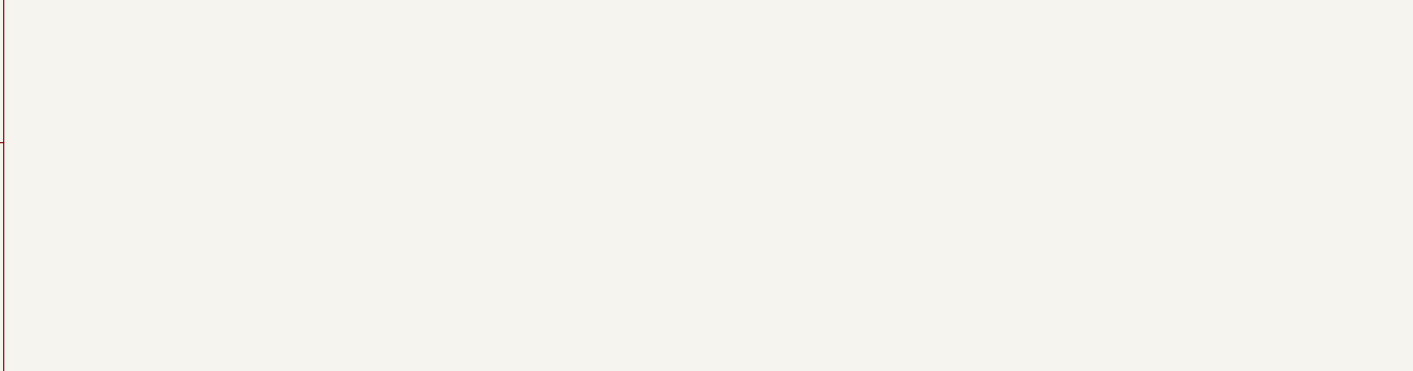
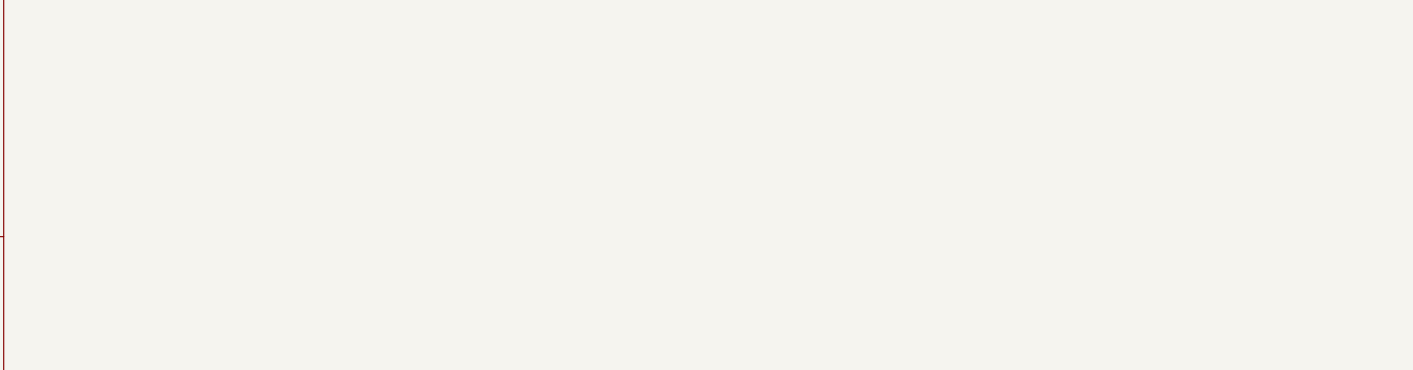
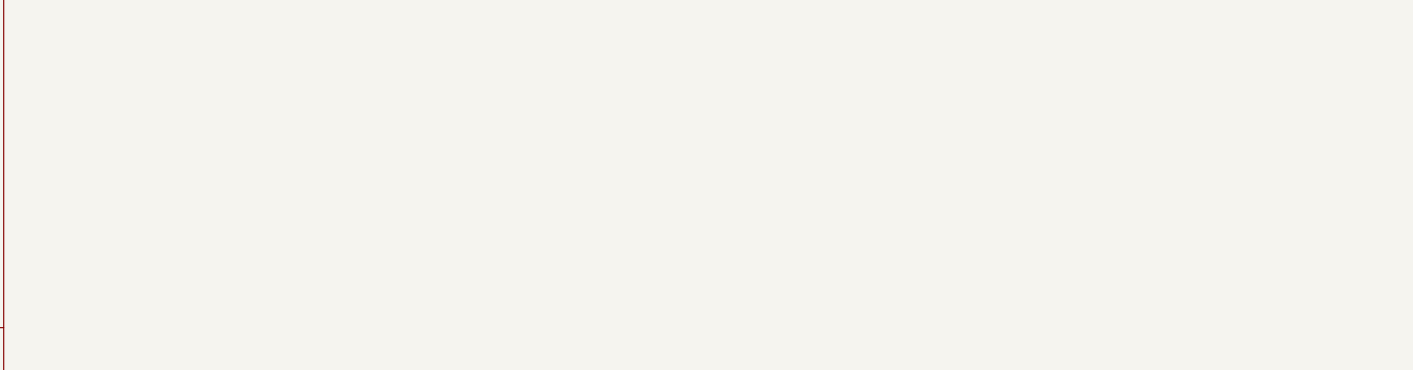
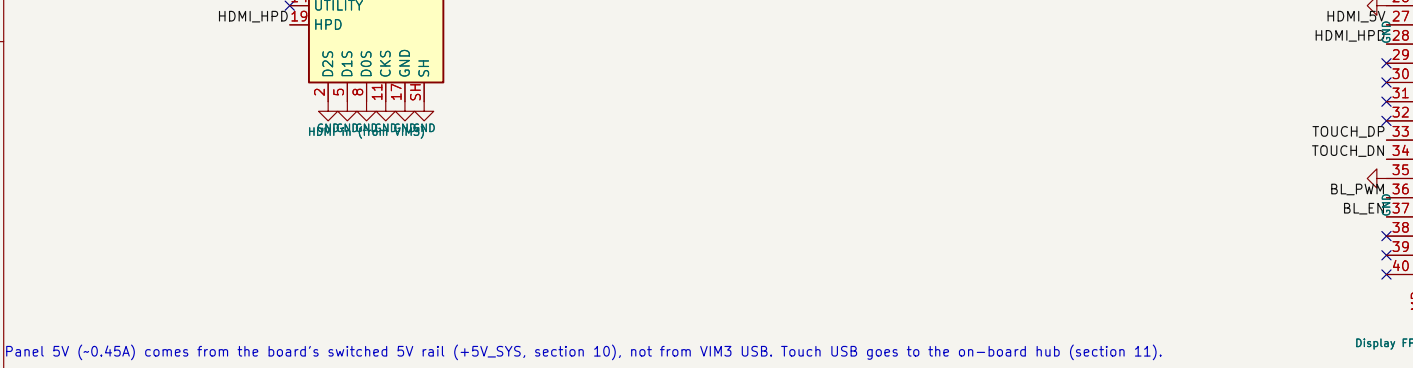
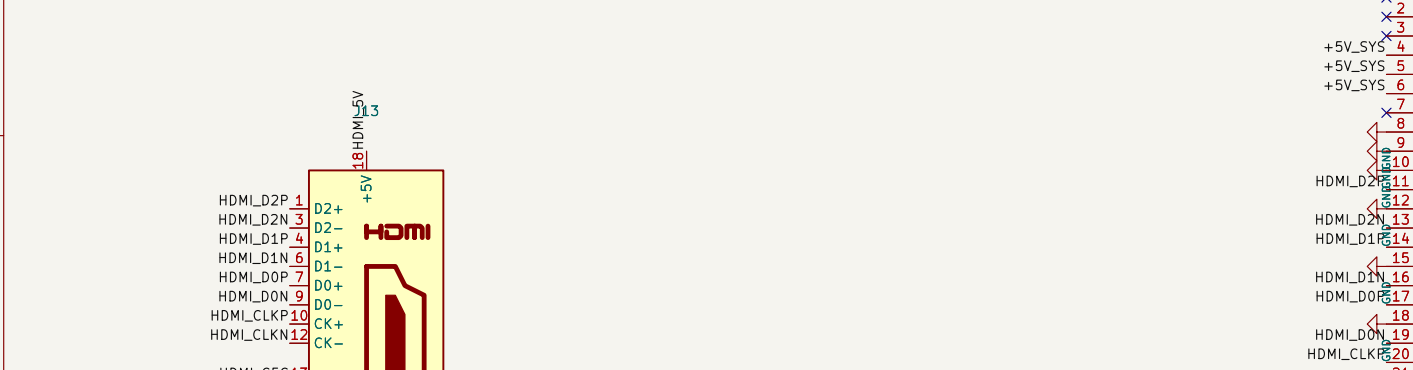
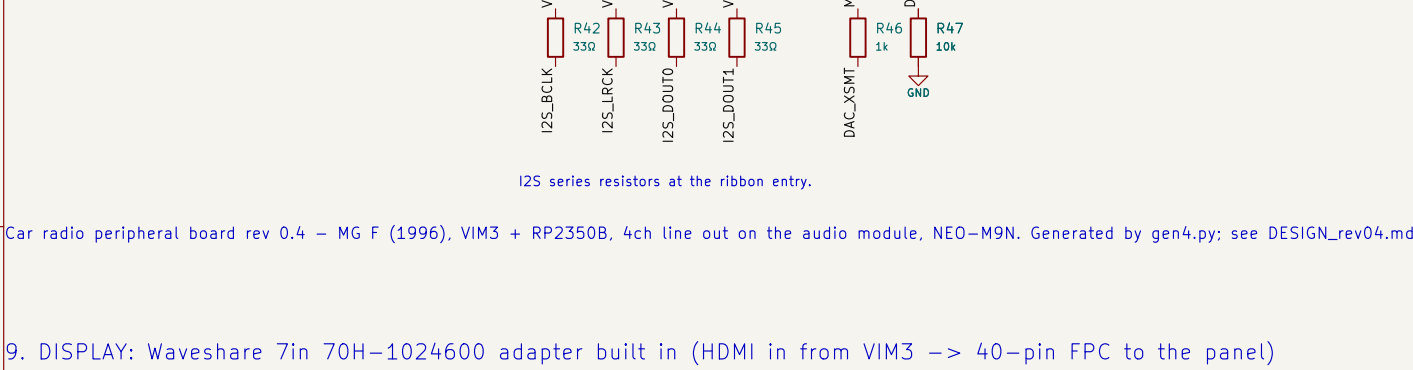
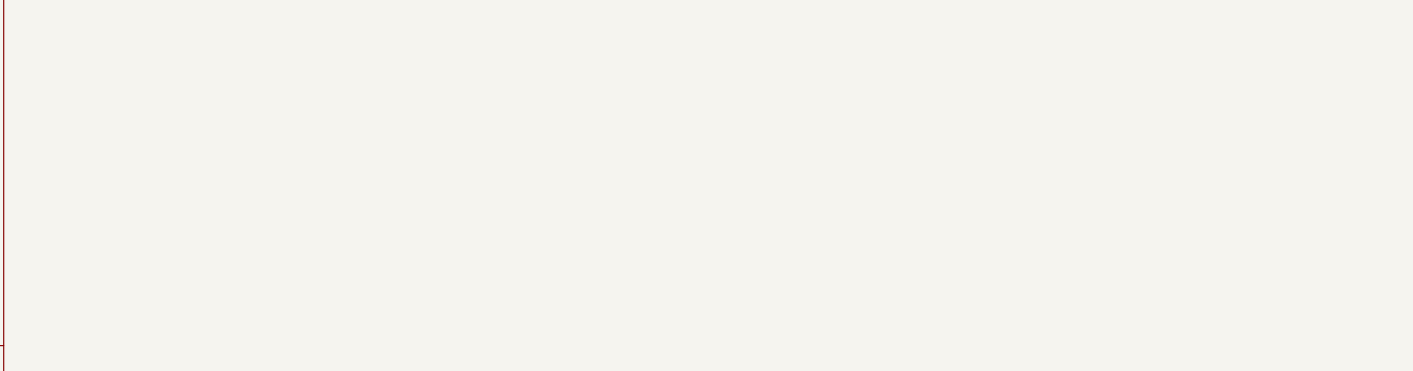
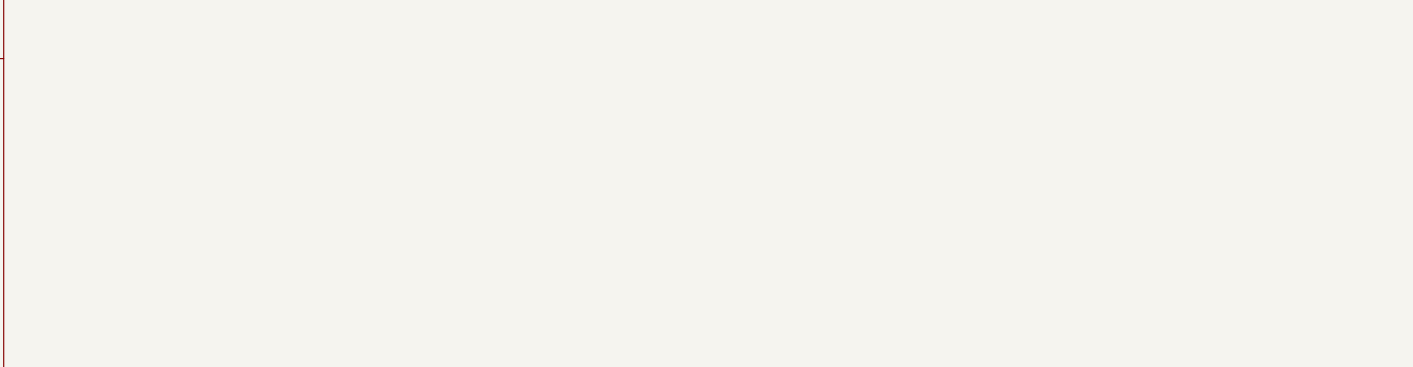
1. ISO 10487 CONNECTOR A (power) + ALWAYS-ON 5V FOR PICCO AND GNSS



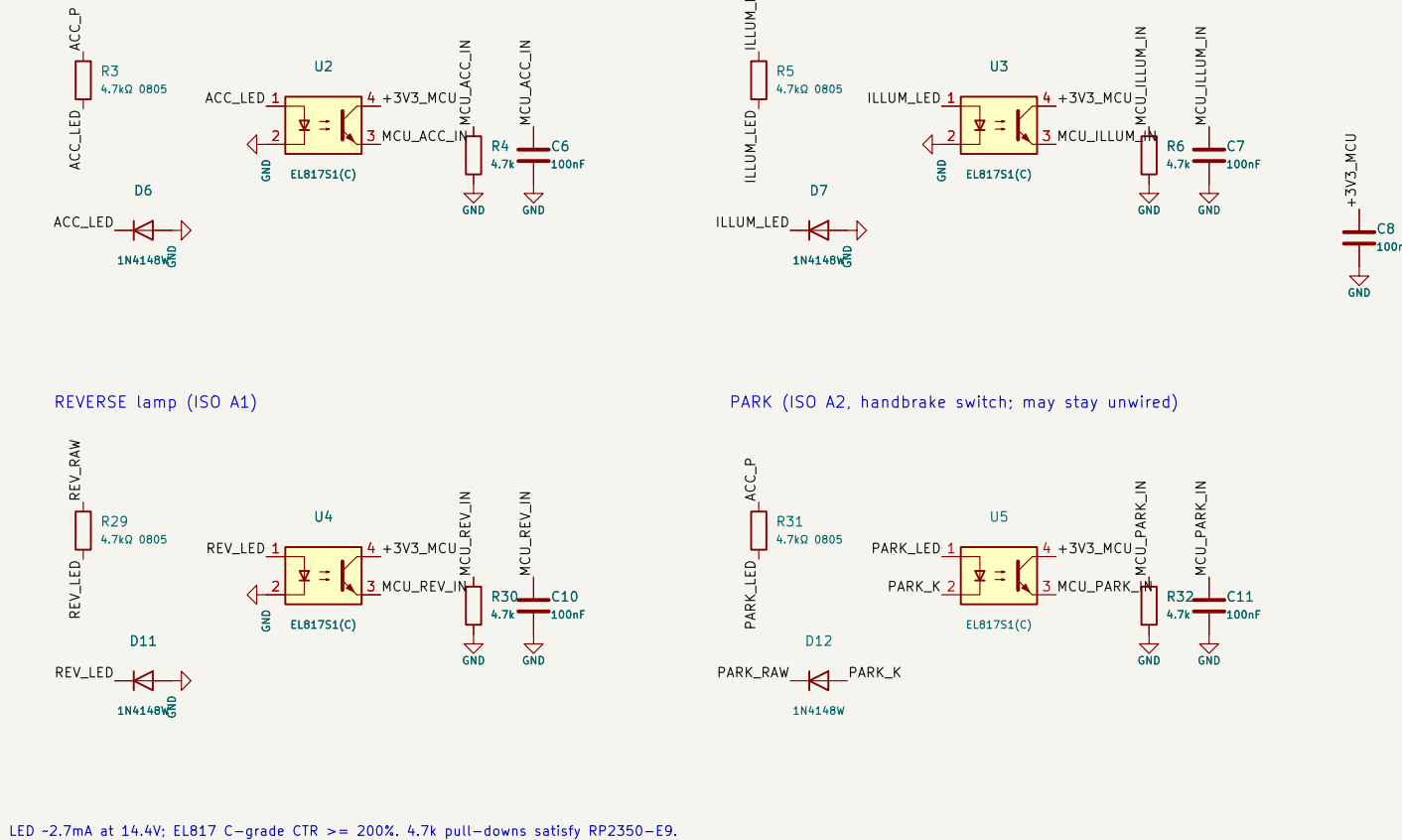
4. MCU HELPERS (MCU core is section 13)



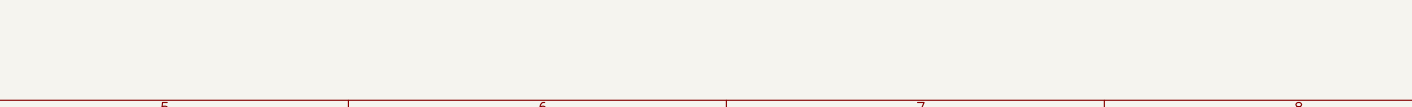
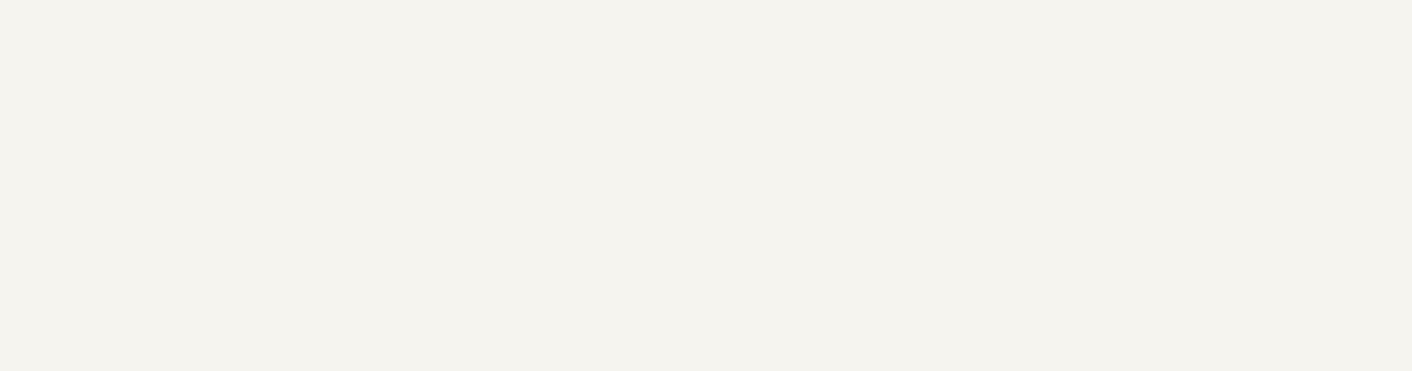
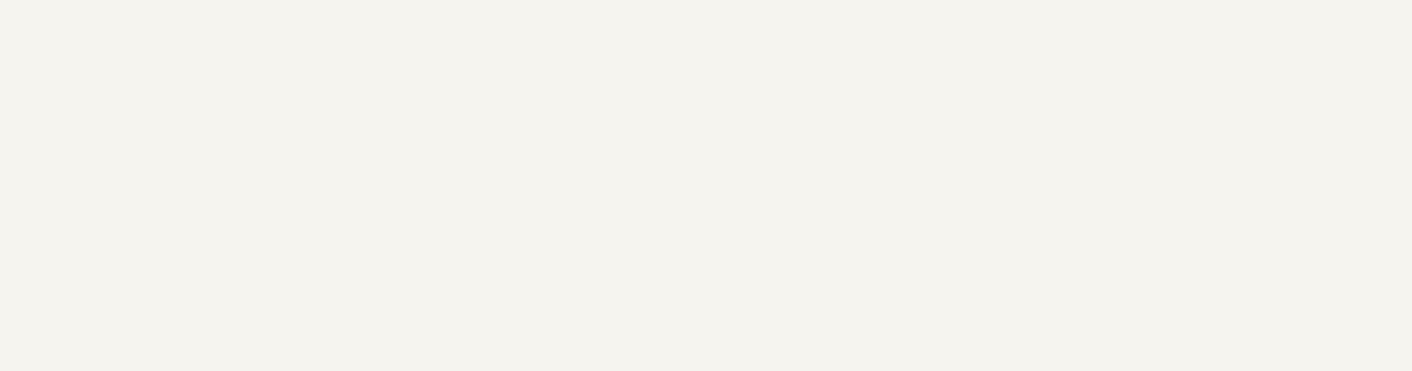
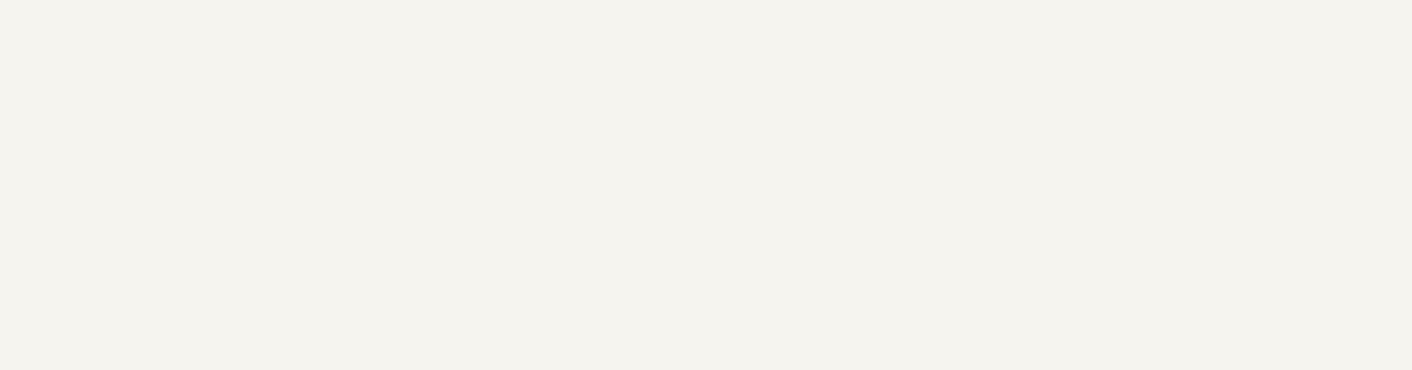
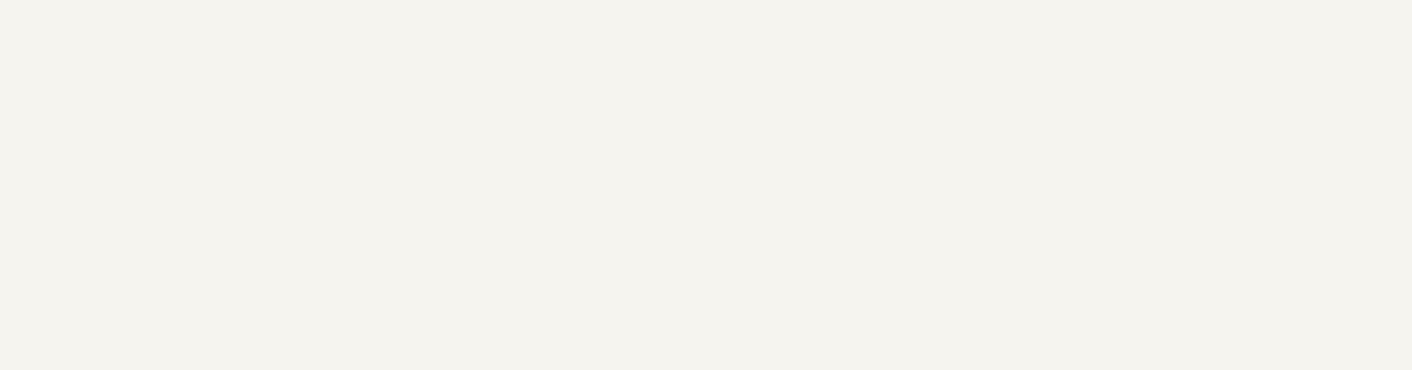
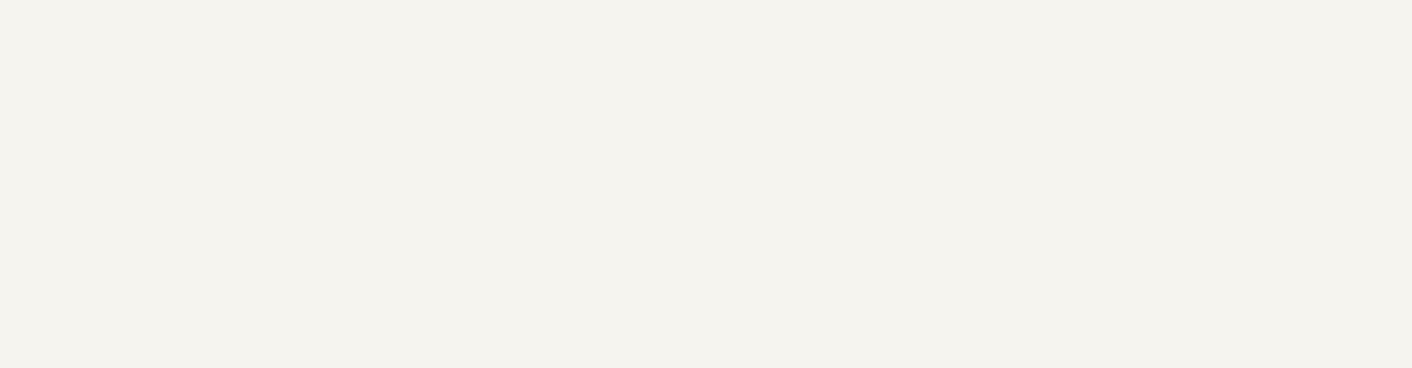
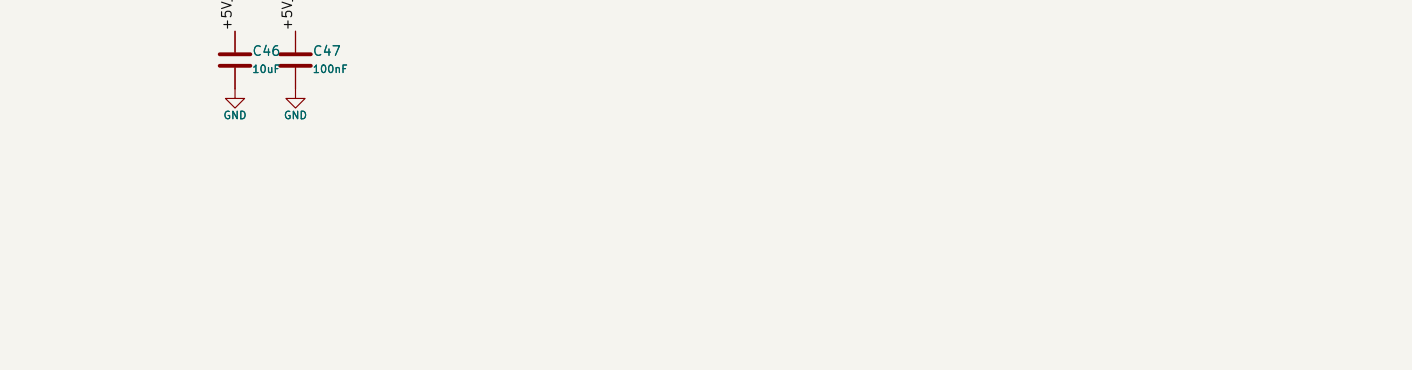
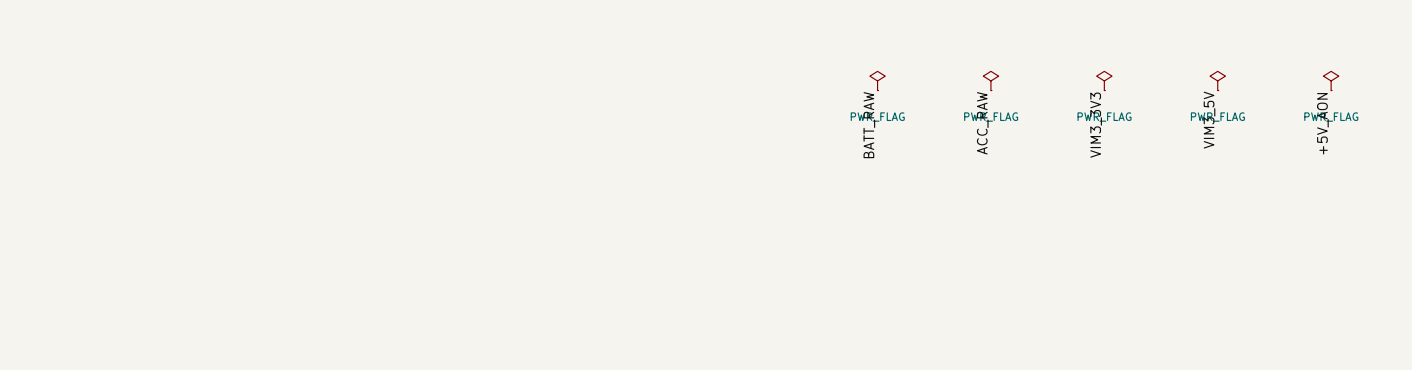
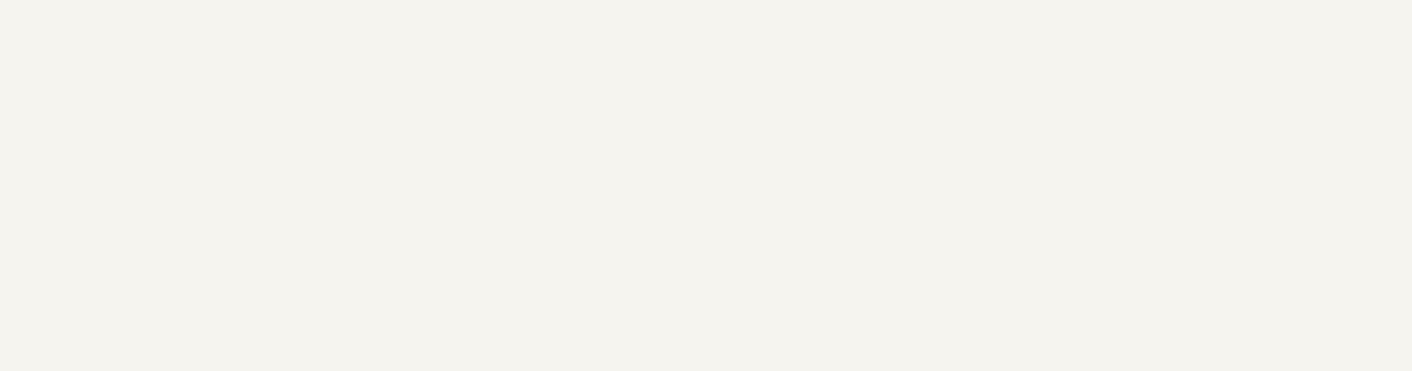
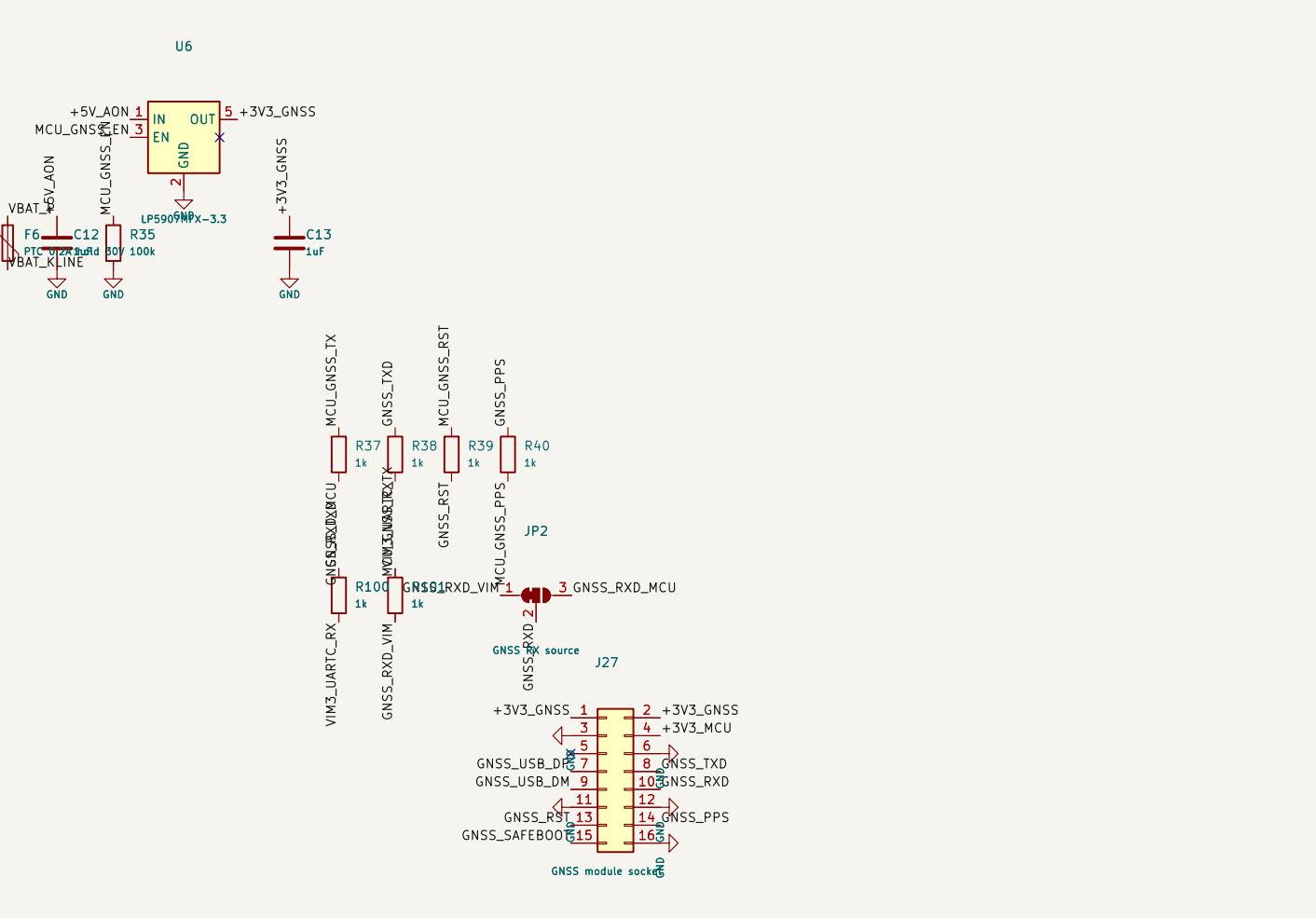
GPIO map see section 13 and DESIGN_rev03.msk.



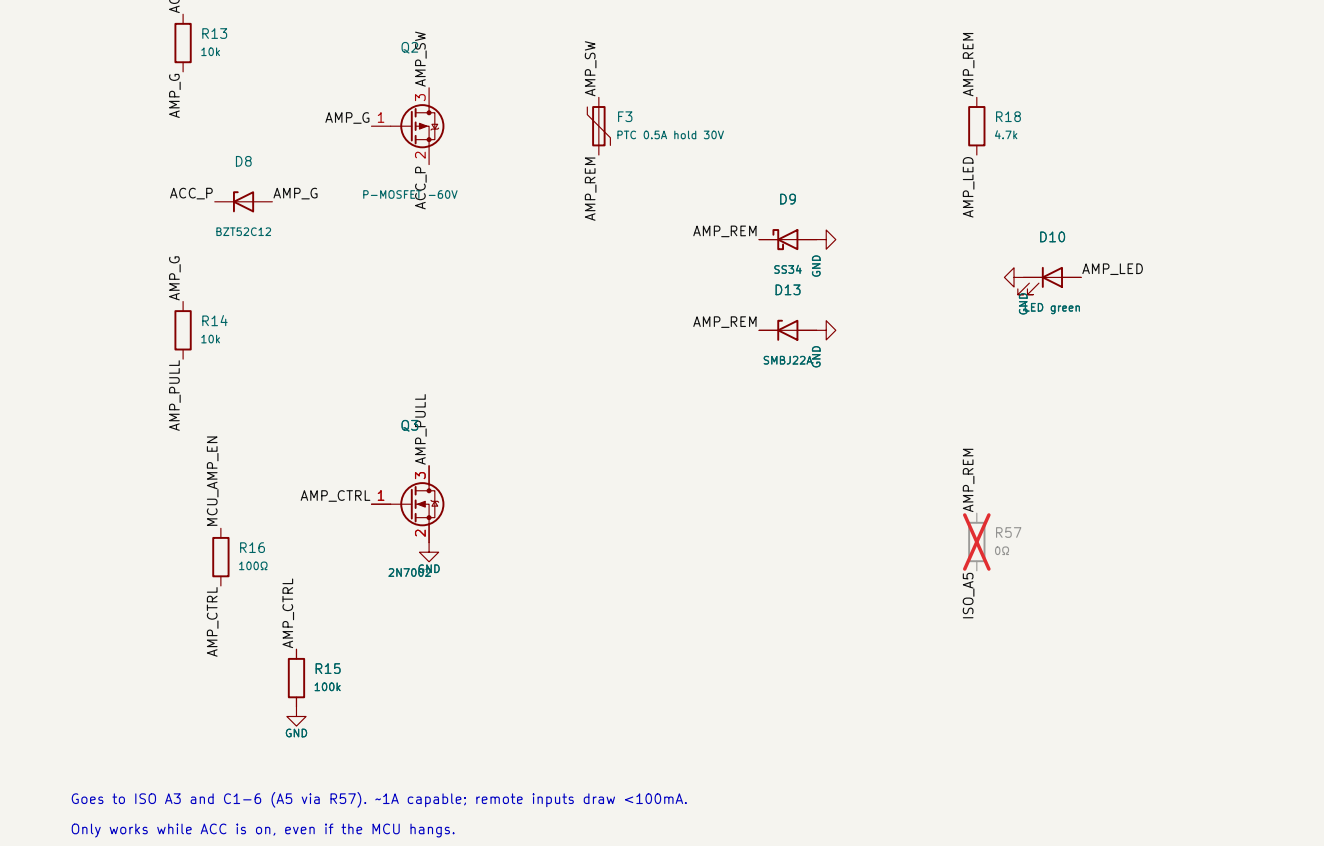
2. VEHICLE INPUTS (opto-isolated, ACC, ILLUM, REVERSE: +12V = on, PARK: switch to ground = on).



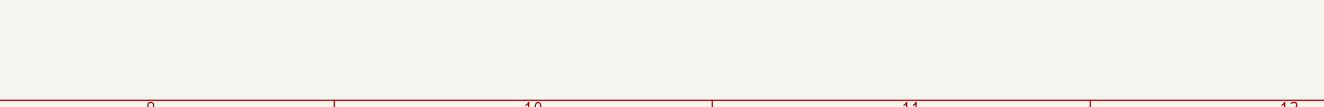
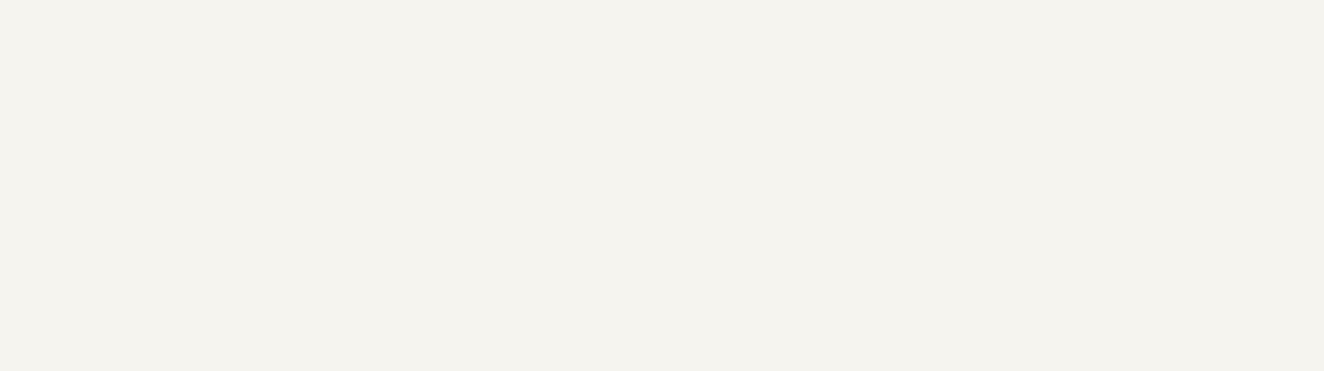
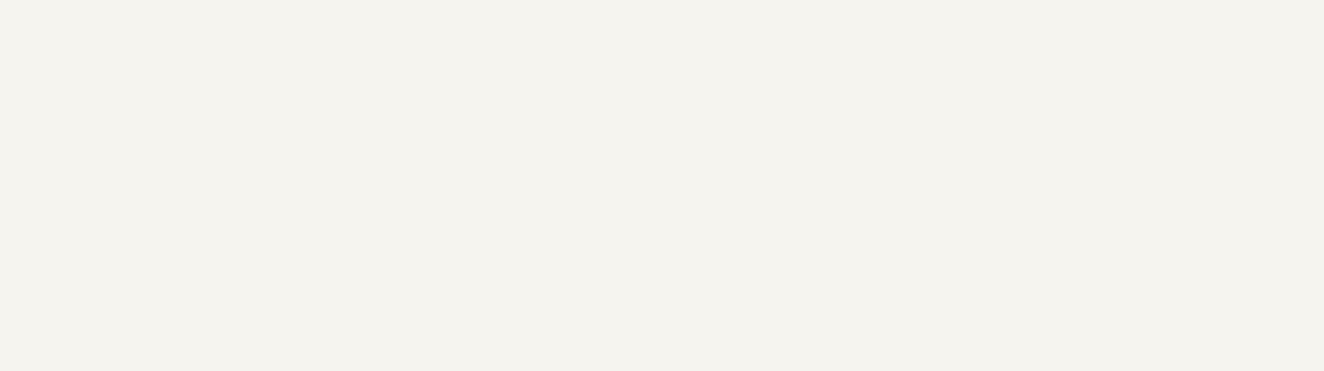
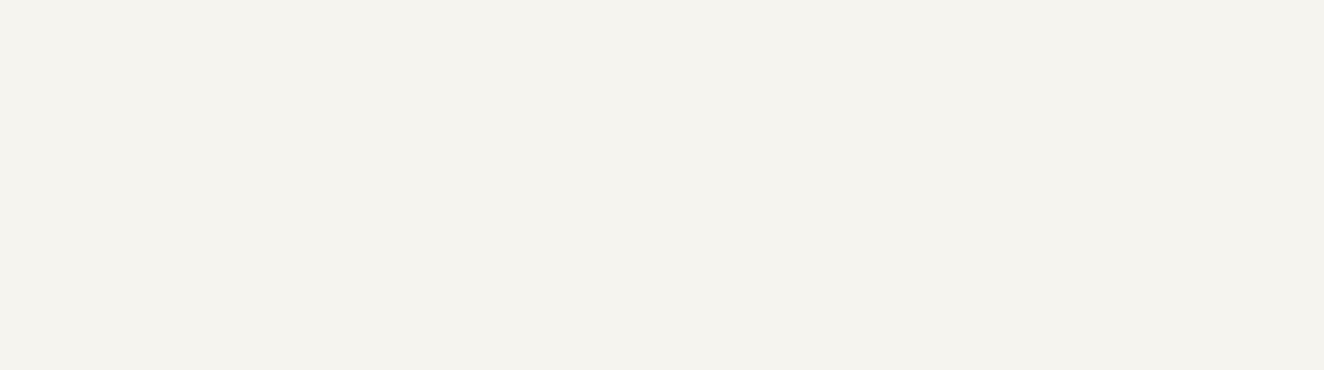
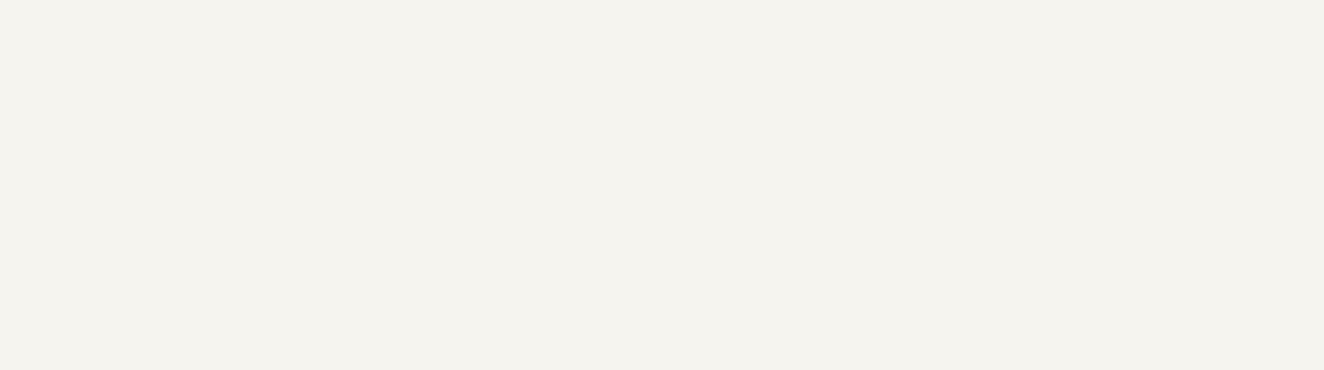
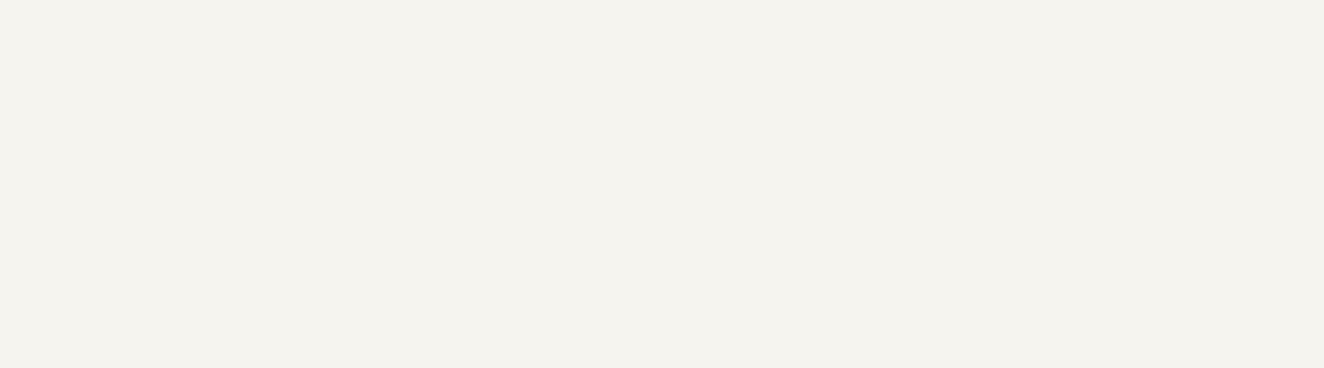
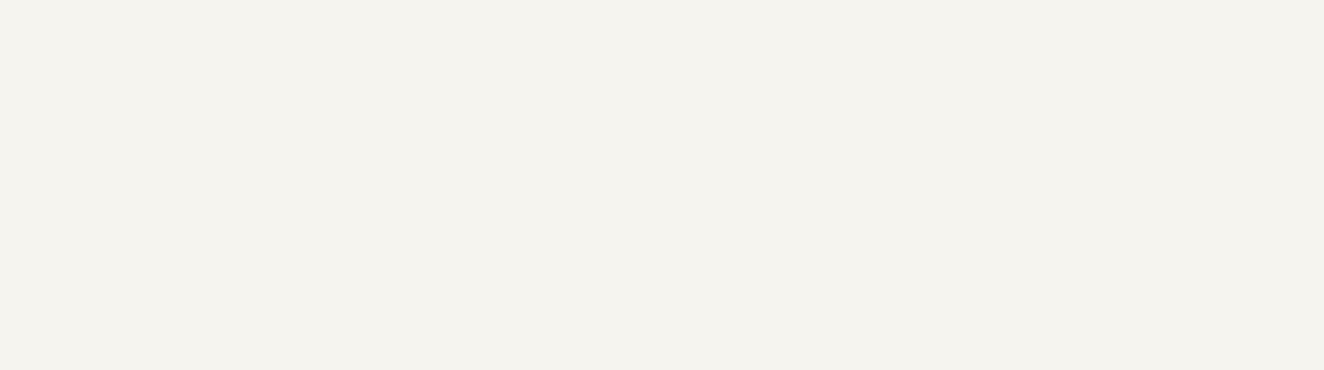
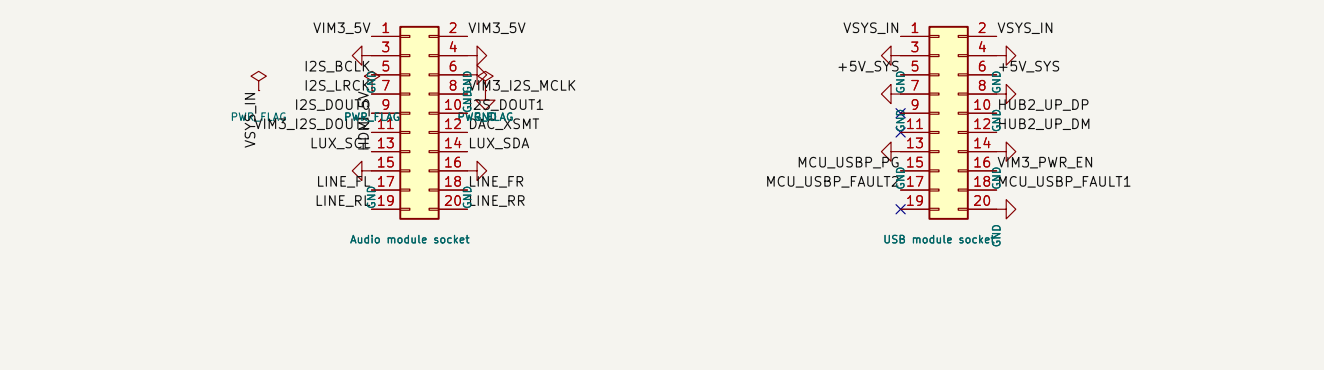
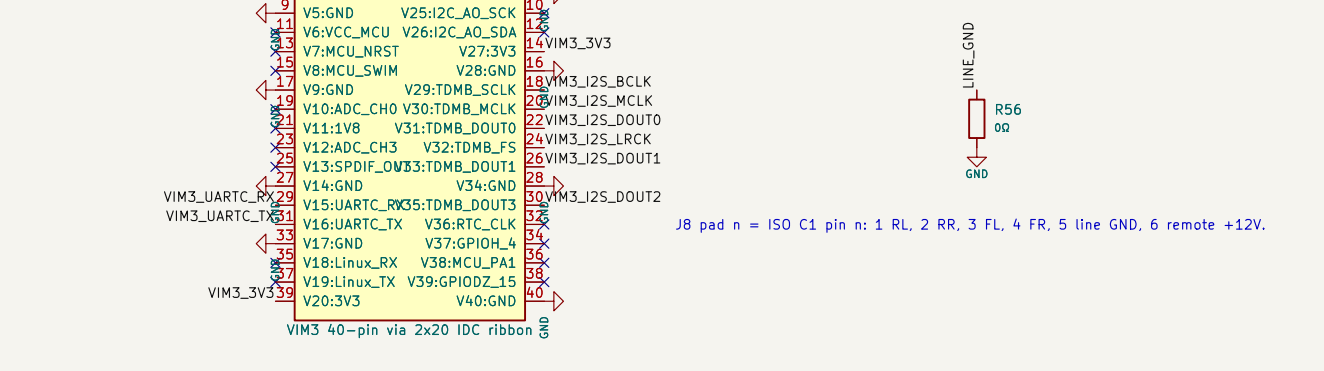
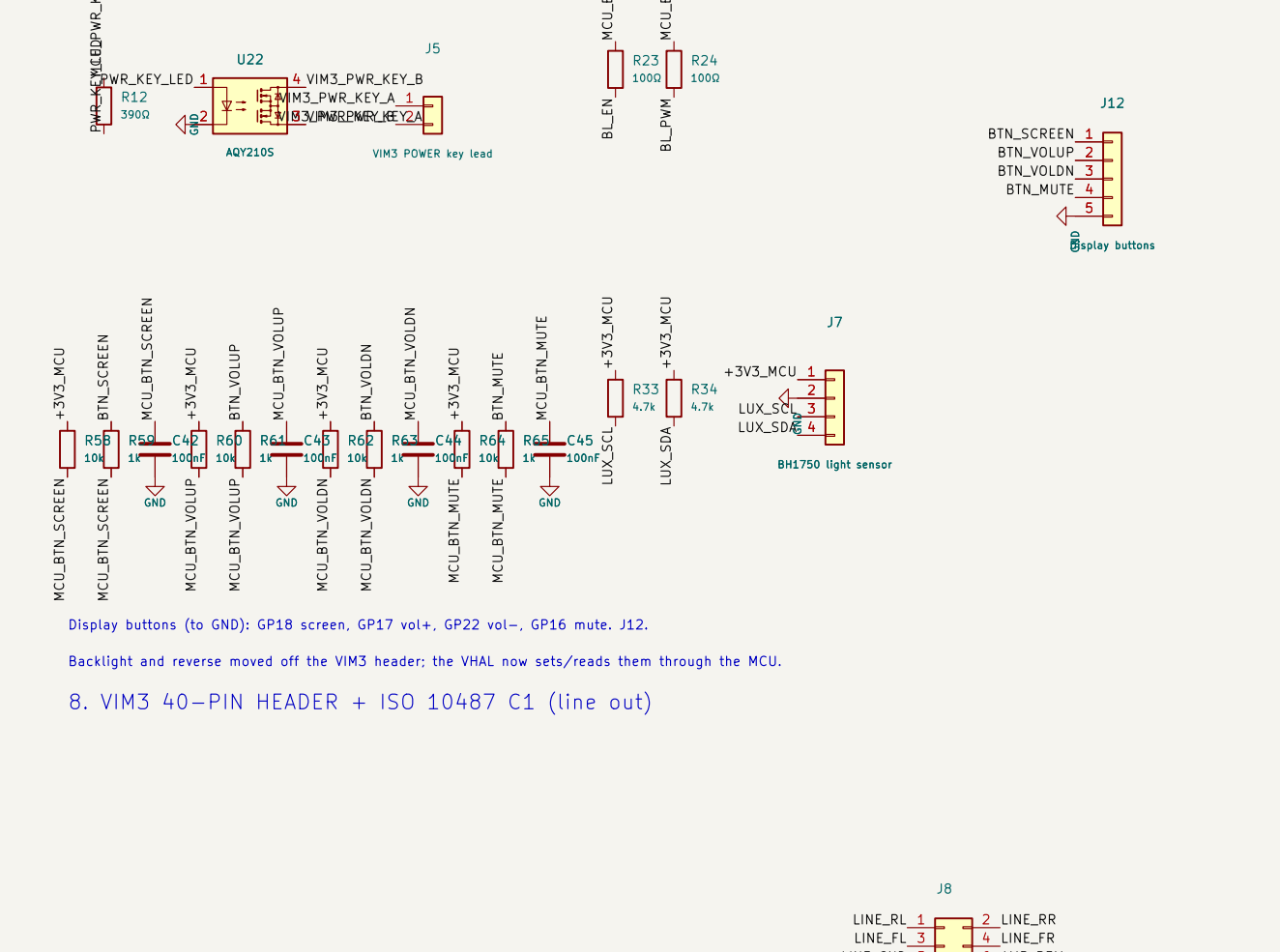
5. GNSS MODULE: u-blox NEO-M9N, active antenna on U.FL (digital to SMA), UART + USB to the main board. Switched 3.3V supply stage VIM3+POWER, B66+BACKLIGHT, LIGHT SENSOR (all from the MCU)



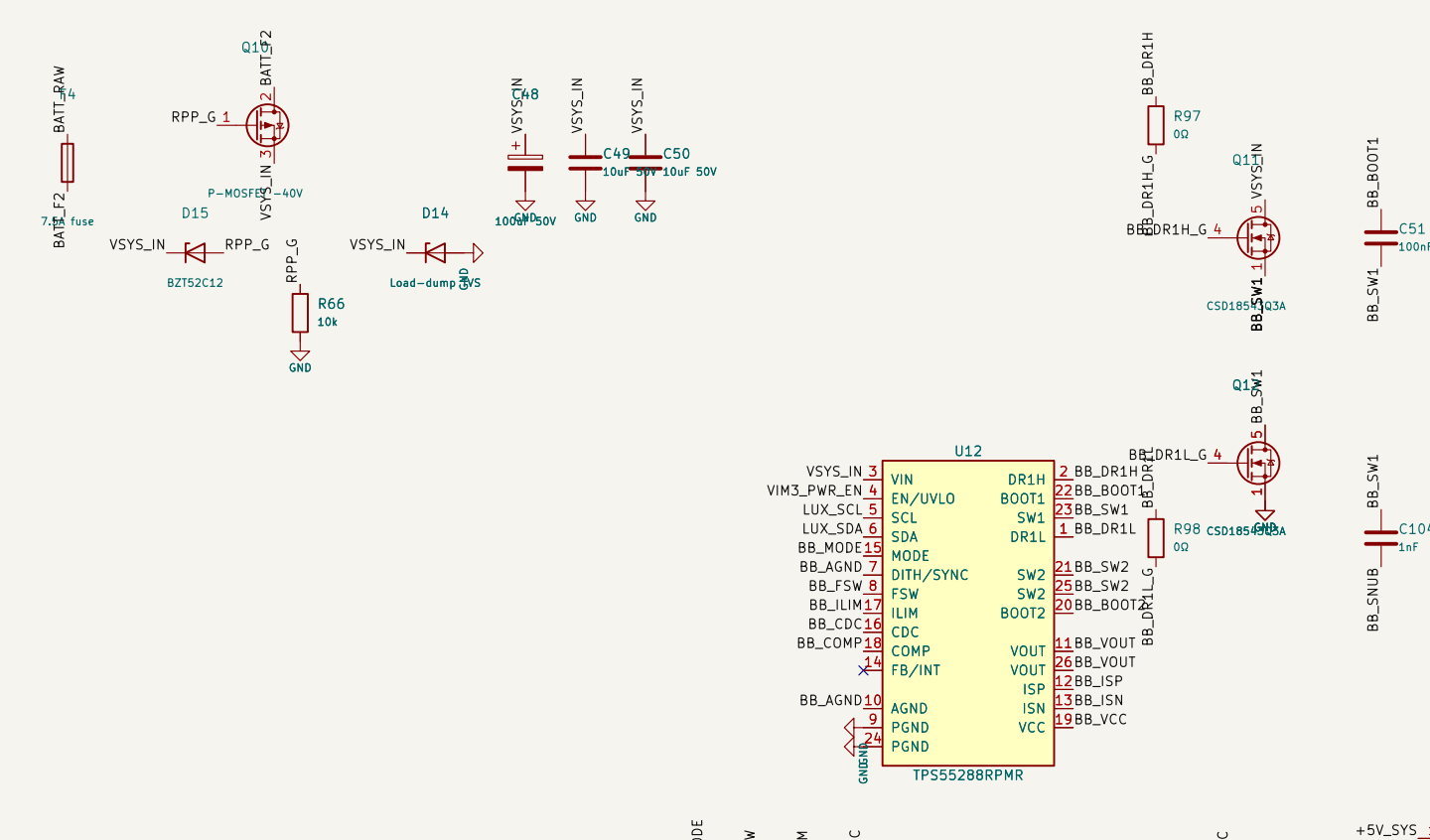
3. AMPLIFIER REMOTE (+12V from ACC, MCU GP20)



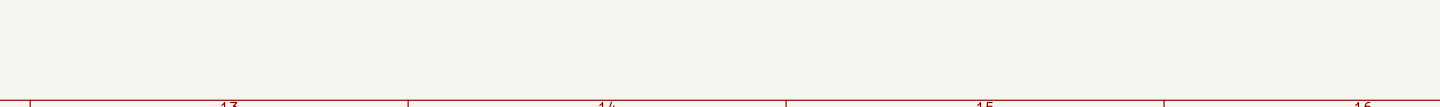
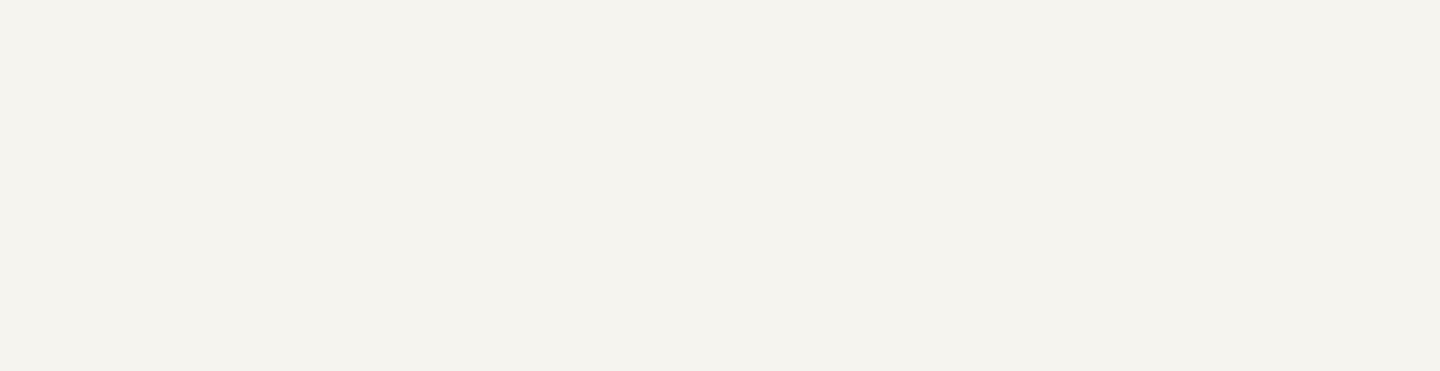
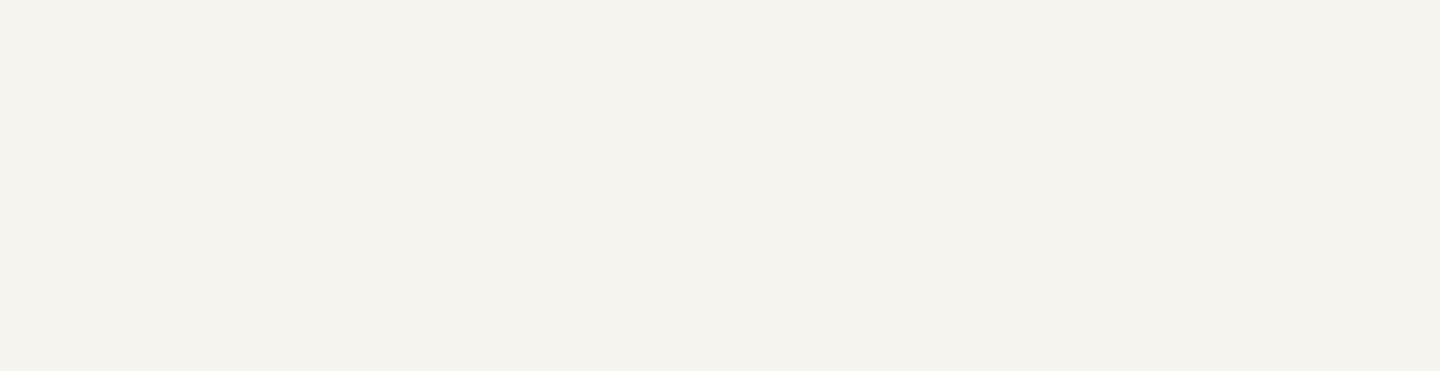
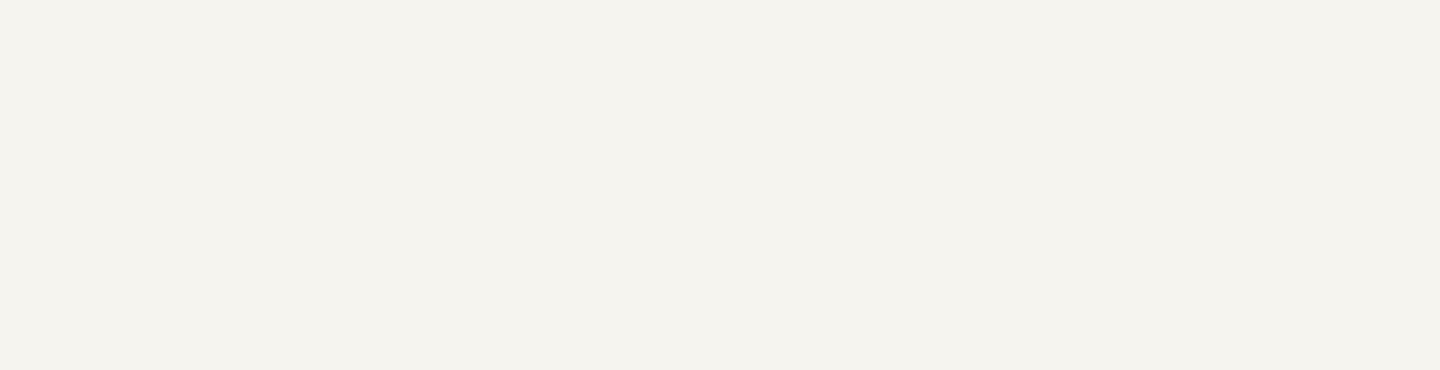
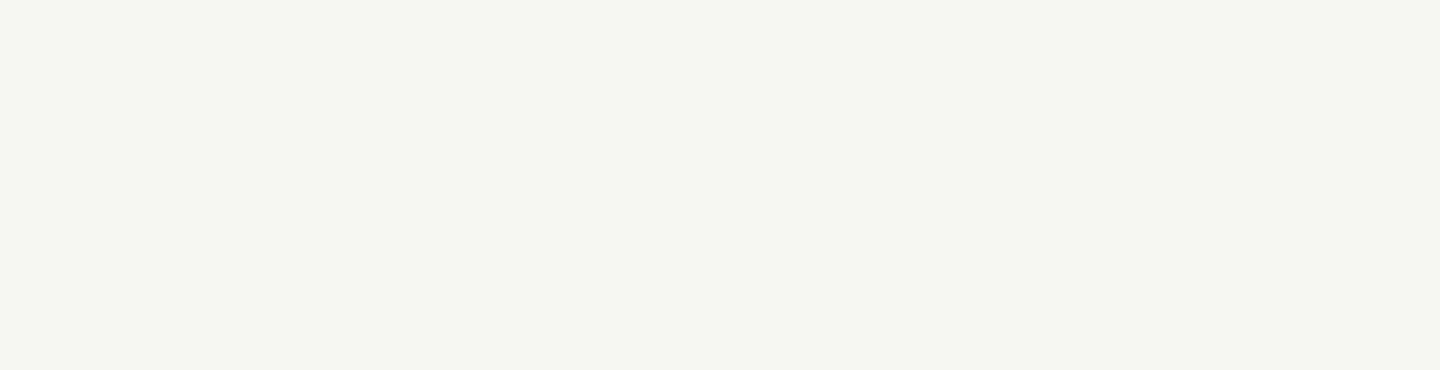
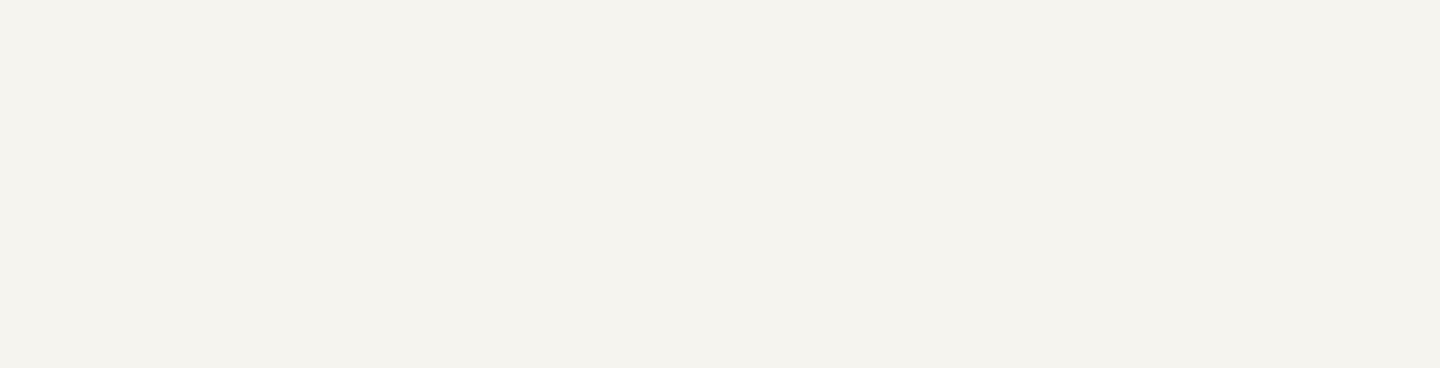
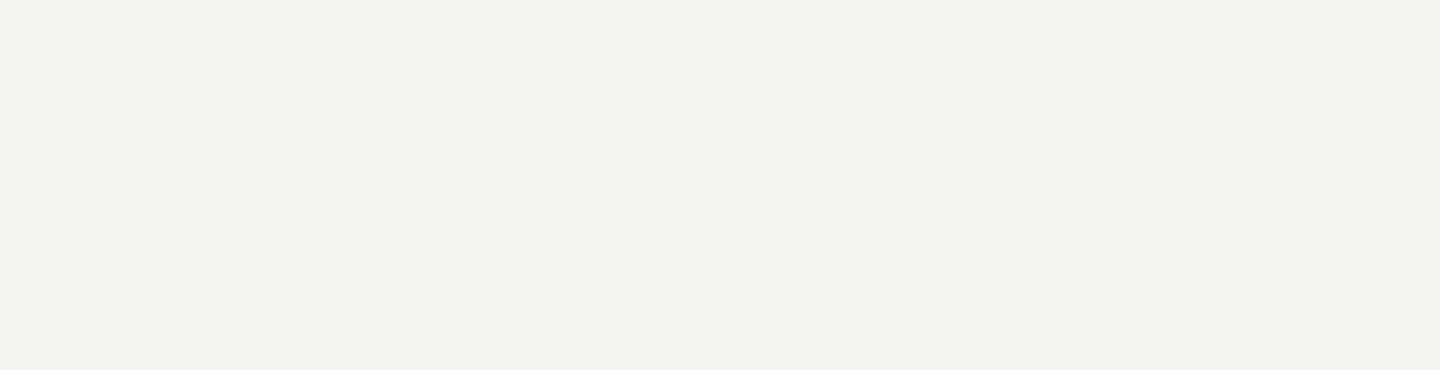
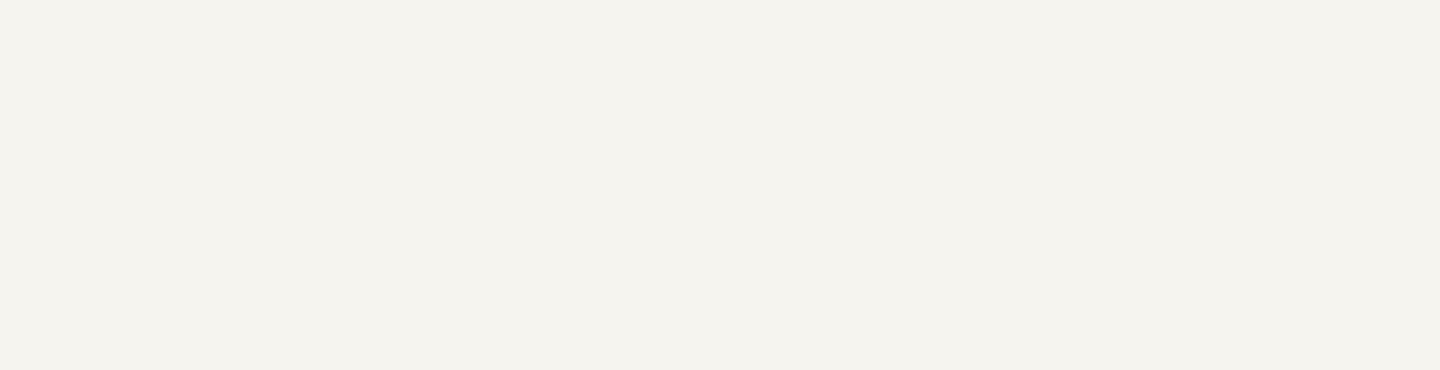
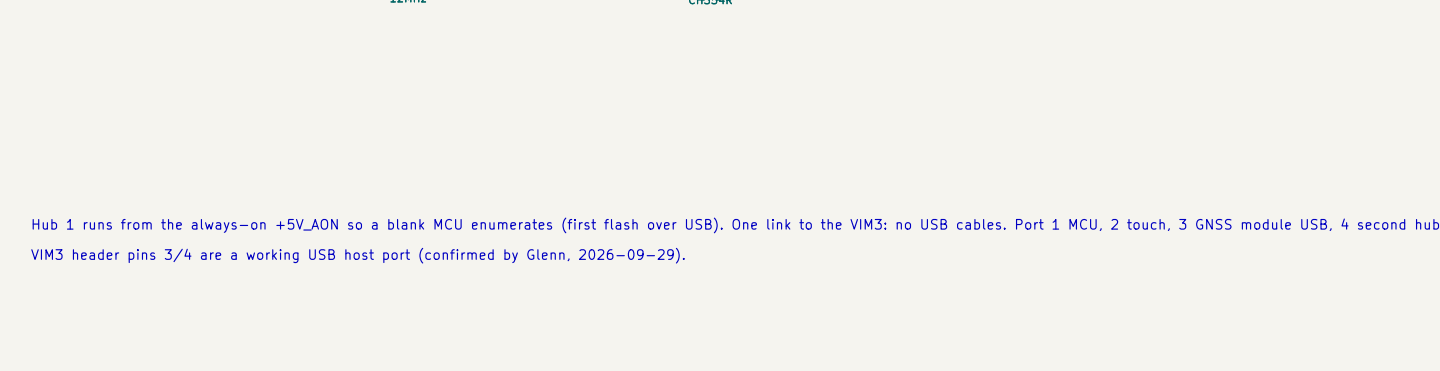
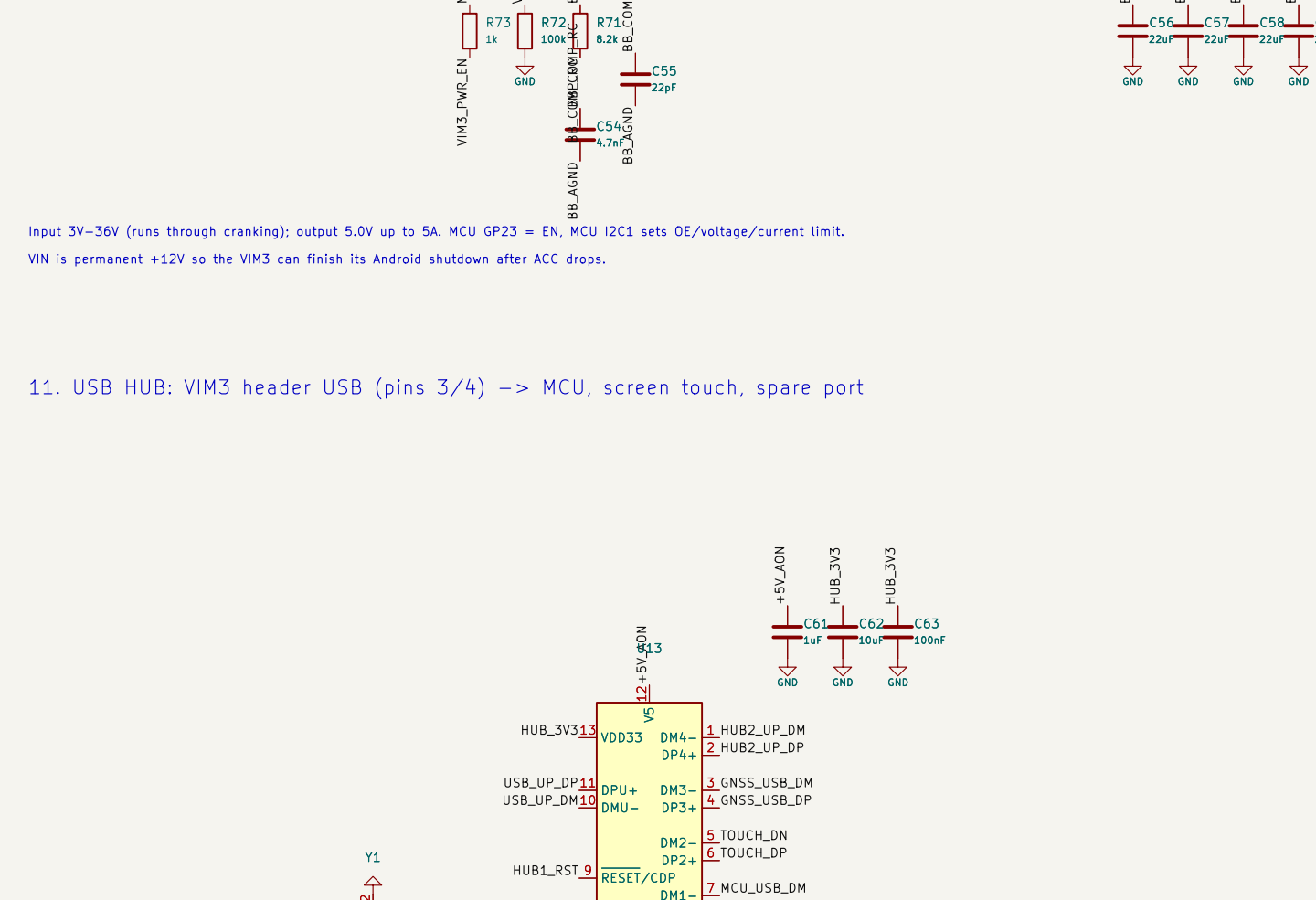
10. 5V/5A SYSTEM SUPPLY FOR VIM3 + SCREEN + HUB (TP555288 buck-boost; rides through cranking and overvoltage)



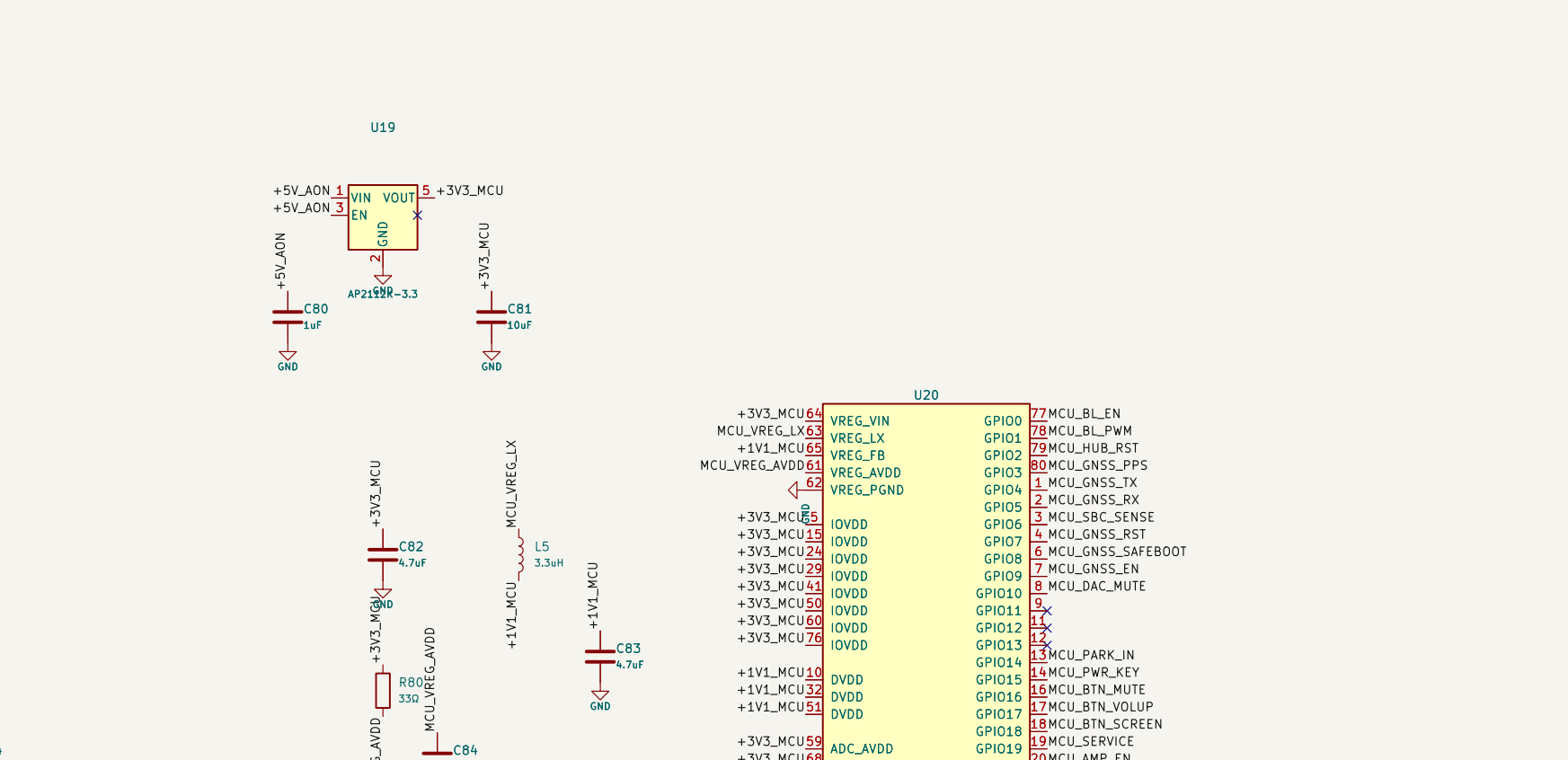
11. USB HUB: VIM3 header USB (pins 3/4) -> MCU, screen touch, spare port



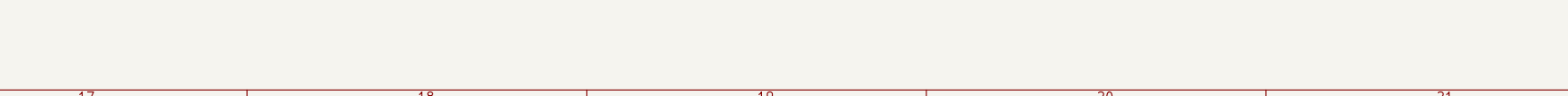
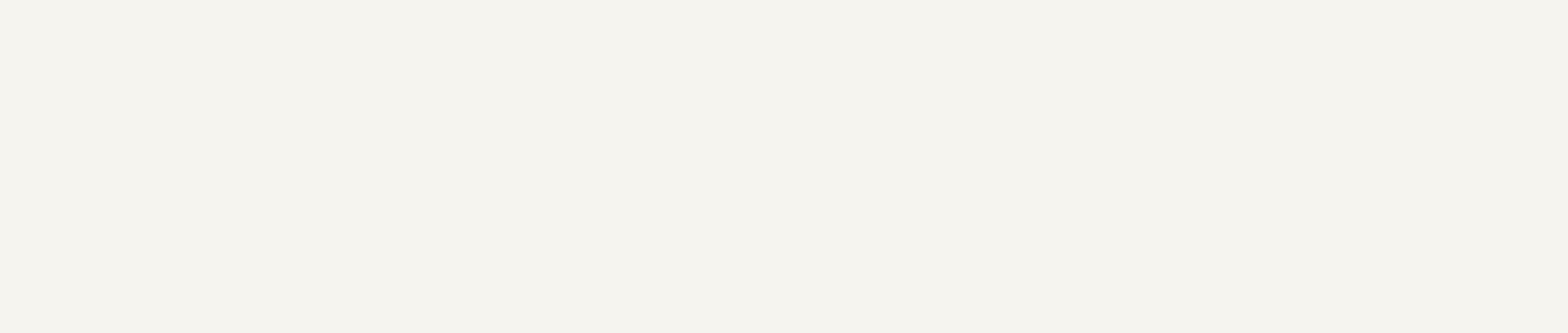
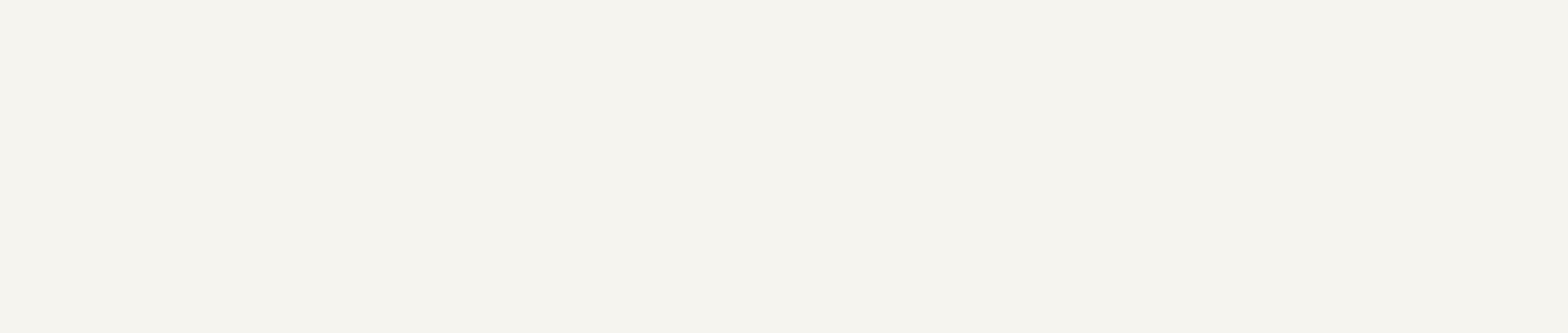
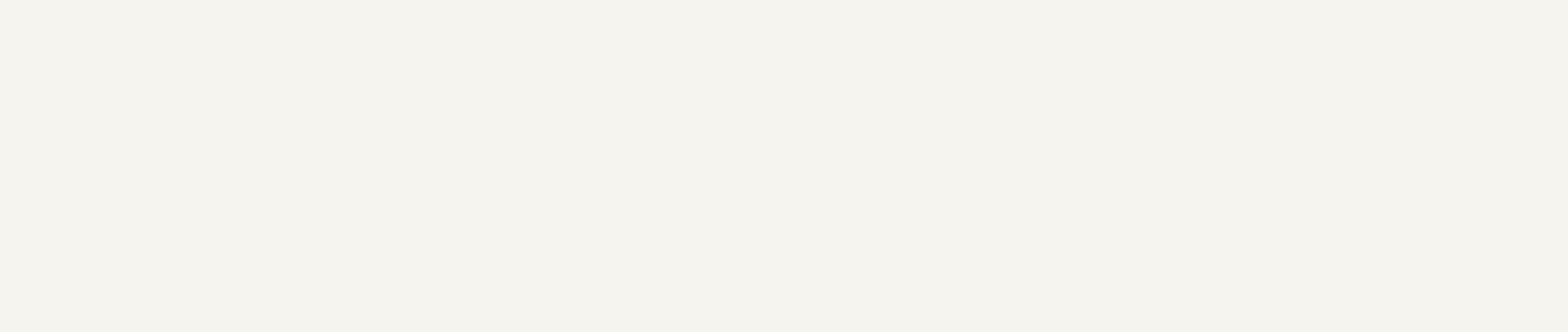
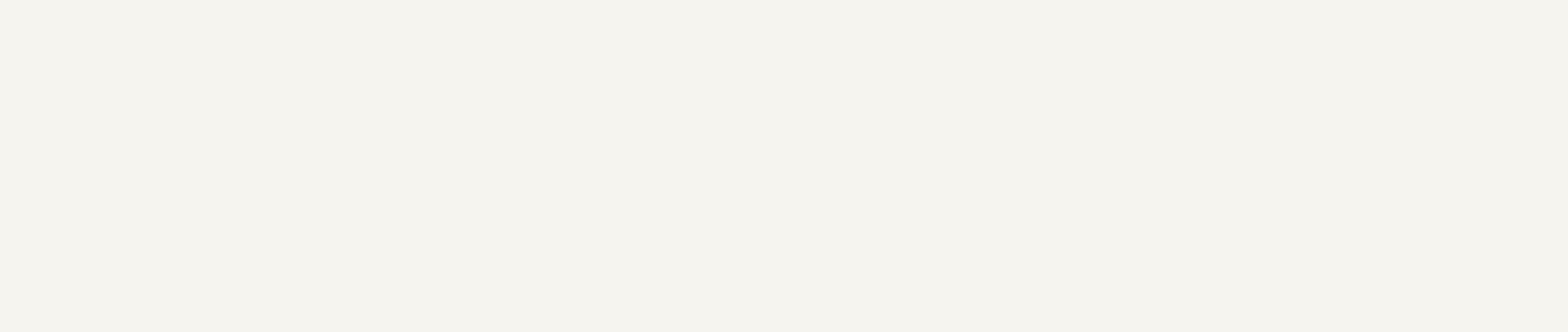
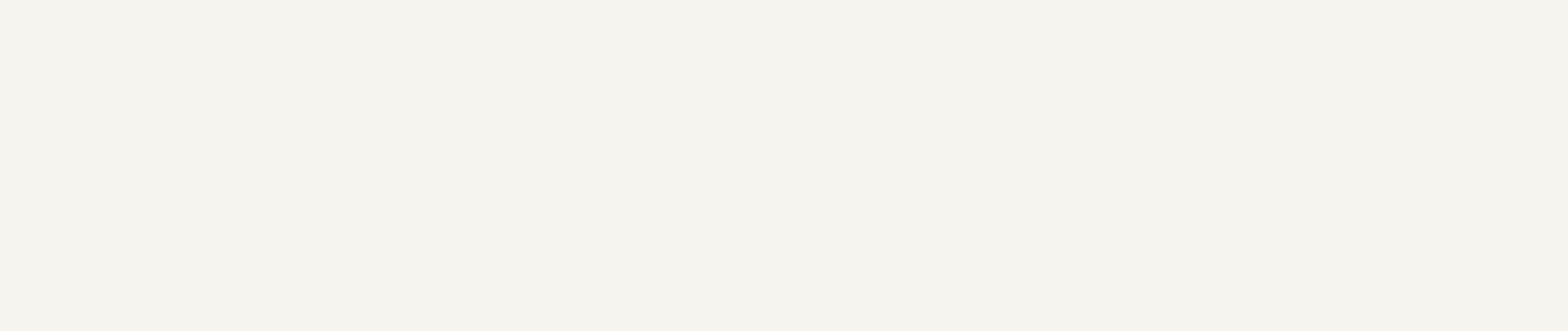
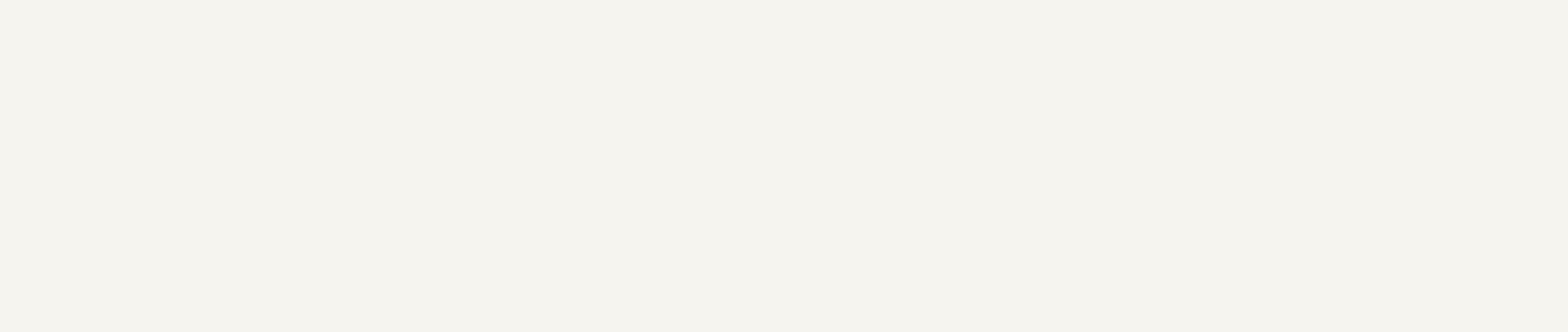
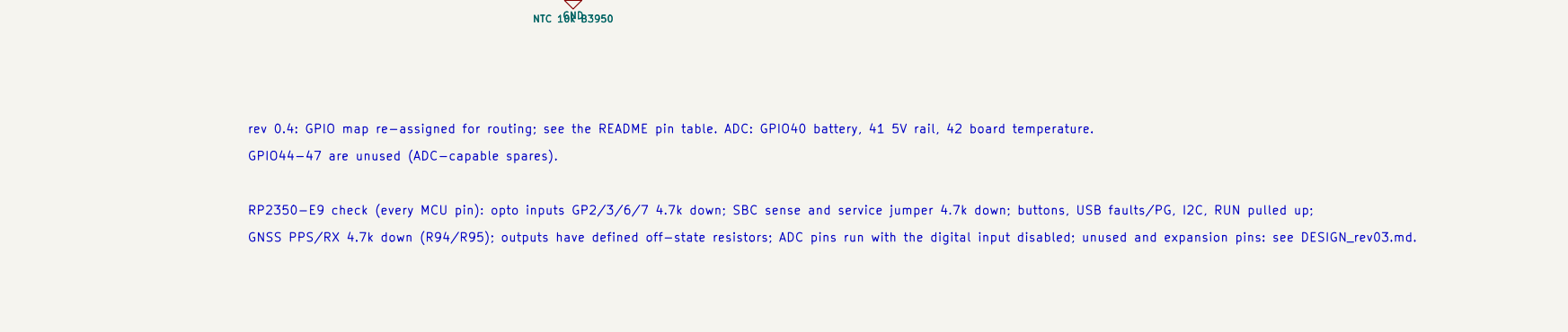
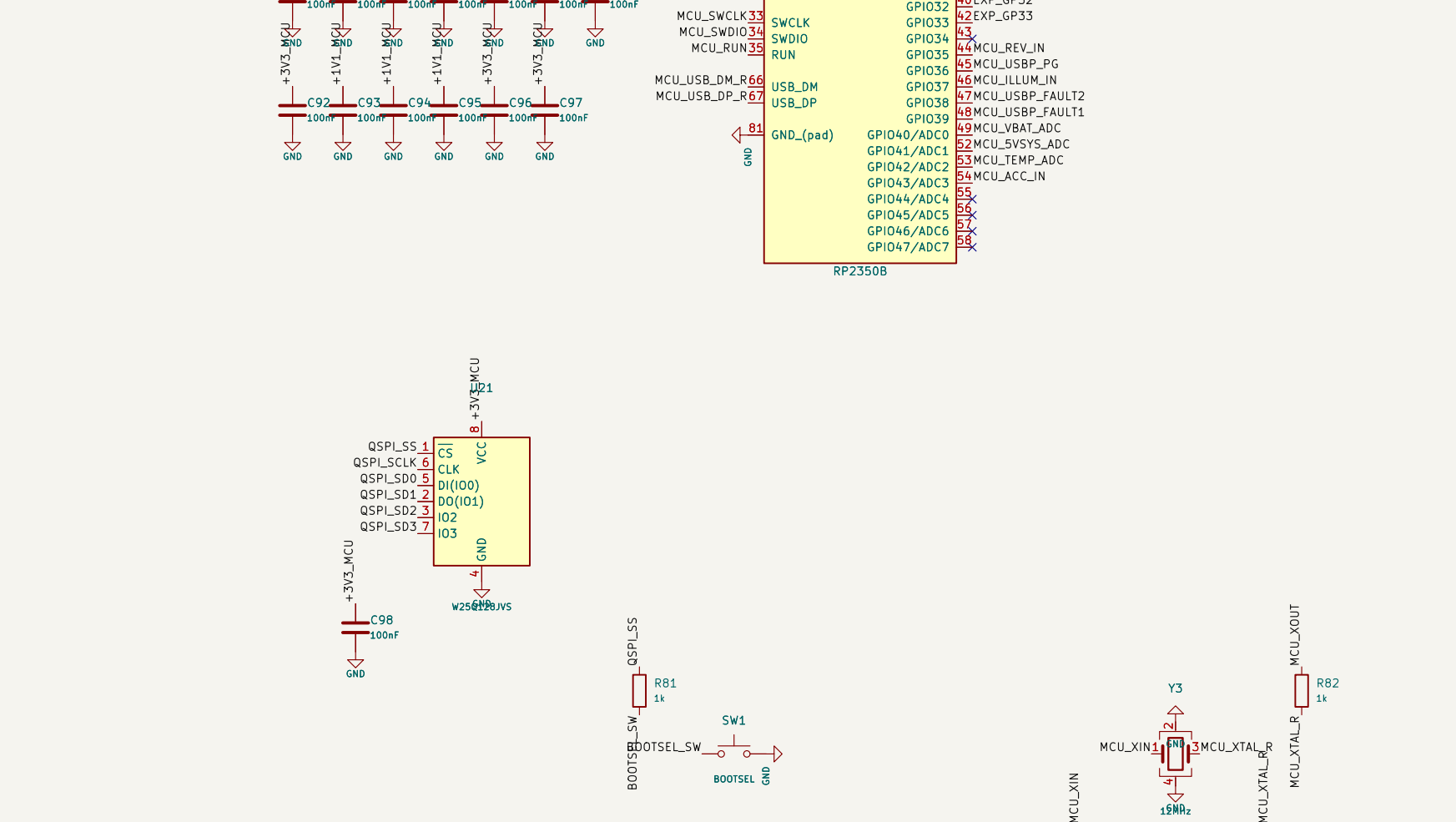
13. MCU: RP2350B (QFN-80) + 16MB QSPI flash + 12MHz crystal + 3.3V LDO, per the Raspberry Pi Pico 2 reference design



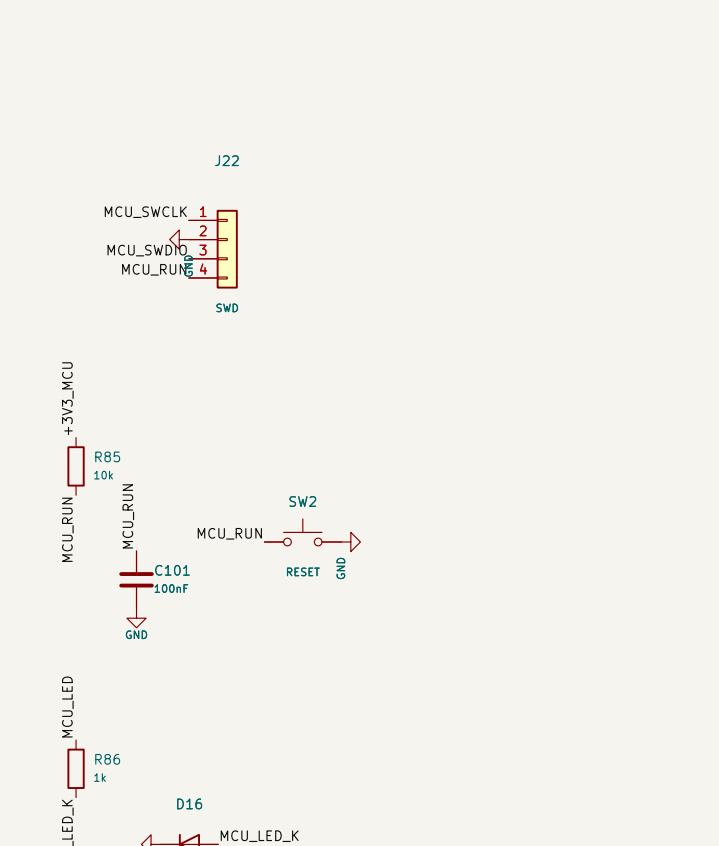
12. DISPLAY: Waveshare Tin 70H-1024x600 adapter built in (HDMI in from VIM3 -> 40-pin FPC to the panel)



14. MCU HELPERS (MCU core is section 13)



15. MCU: RP2350B (QFN-80) + 16MB QSPI flash + 12MHz crystal + 3.3V LDO, per the Raspberry Pi Pico 2 reference design



16. MCU: RP2350B (QFN-80) + 16MB QSPI flash + 12MHz crystal + 3.3V LDO, per the Raspberry Pi Pico 2 reference design

